

Traveling phase waves in asymmetric networks of noisy chaotic attractors

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We explore identical Rössler systems organized into two equally sized groups, among which differing positive and negative in- and out-coupling strengths are allowed. With this asymmetric coupling, we analyze patterns in the phase dynamics that coexist with chaotic amplitudes. We specifically investigate traveling phase waves where the oscillators settle on a new rhythm different from their own. We show that these waves are possible even without coherence in the phase angles. It is further demonstrated that the emergence of these *incoherent traveling waves* depends on the type of coupling, not on the individual dynamics of the Rössler systems. Together with the study of noise effects, our results suggest a promising new avenue toward the interplay of chaotic, noisy, coherent, and incoherent collective dynamics.

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I. INTRODUCTION

Discovering that coupled chaotic oscillators can synchronize their phase angles [1] marked a milestone in the study of collective synchronization. The fascination was due to the fact that the amplitude variables remained chaotic and uncorrelated. This phenomenon has now found general interest in many practical applications [2].

In this paper we show that even more subtle collective rhythms are possible in chaotic systems. It is demonstrated that the combination of positive and negative coupling strengths among coupled chaotic oscillators yields traveling phase waves (TW). In such a state, the oscillators settle on a new rhythm different from their own, i.e., the mean ensemble frequency and the locking frequency (when definable) of the bulk of oscillators do not equal anymore the natural one. Remarkably, this behavior is possible even in absence of phase synchronization. We speak of *incoherent traveling waves* when describing this collective state in which spontaneous drifts are observed with incoherent oscillations.

The existence of TWs has remained unnoticed in the phase synchronization of chaotic systems. Our work hence provides a significant broadening for the emergence of those collective phenomena, encompassing chaotic oscillatory behavior with coupled phase-amplitude dynamics. In particular, we investigate two different formulations for the coupling setup between the oscillators. First, in the *weighted x -coupled model*, we consider Rössler oscillators [3] with funnel and phase-coherent attractors coupled through the x coordinate. In this case, besides the occurrence of π states, where the oscillators split into two clusters separated by a phase lag of π , a novel dynamical state is reported. In contrast to phase oscillator systems [4–6], it is found that a noticeable spontaneous drift in the frequencies occurs only when the system is in the incoherent state. These properties define the incoherent TW state. While there is a similarity with

the synchronization of phase-coupled systems regarding the occurrence of incoherent and partially synchronized states, we show that chaotic oscillators diffusively coupled via repulsive and attractive interactions exhibit a strikingly different dynamics with respect to variations in the collective frequency.

Second, in the *phase-coupled model*, oscillators operating in the phase-coherent regime interact purely through the sine of the phase differences. In contrast with the weighted x -coupled model, it is shown that TWs among phase-coupled Rössler oscillators only occur if the system converges to partially synchronized states in which the subpopulations evolve separated by a constant phase lag between 0 and π . Even though the phase angles depend on the chaotic amplitudes, we are able to estimate the onset of partial synchronization and the transitions to π states using calculations based on the paradigmatic Kuramoto model. These findings bridge the gap to phase models providing at the same time a rounded picture on the emergence of traveling phase waves.

II. RÖSSLER SYSTEMS WITH DIFFUSIVE COUPLING

Let us consider first a coupling setup where the oscillators interact through the x coordinate via a weighted network [8,9] defined by the matrix $\mathbf{W} = [w_{ij}]$ as follows:

$$\begin{aligned}\dot{x}_i &= -\omega_0 y_i - z_i + \frac{1}{N} \sum_{j=1}^N w_{ij} (x_j - x_i) + \xi_i(t), \\ \dot{y}_i &= \omega_0 x_i + a y_i, \quad \dot{z}_i = b + z_i (x_i - c).\end{aligned}\tag{1}$$

The phases can be defined as $\phi_i^H(t) = \arctan[x_i^H(t)/x_i(t)]$, with x_i^H being the Hilbert transform of x_i , $x_i^H(t) = \pi^{-1} \int_{-\infty}^{\infty} x_i(\tau)/(t - \tau) d\tau$, where the integral is taken in the sense of the Cauchy principal value [10]. The terms $\xi_i(t)$ correspond to time-dependent disorder modeled as Gaussian white noise satisfying $\langle \xi_i(t) \rangle = 0$, $\langle \xi_i(t) \xi_j(t') \rangle = 2D \delta_{ij} \delta(t - t')$, where D is the noise strength.

The coupling between the oscillators is mediated through the weights w_{ij} , which are henceforth given by $w_{ij} = K_i G_j$. In effect, to each oscillator a pair of couplings (K_i, G_i)

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is associated. Coupling strength K_i is responsible for how strongly node i responds to the interaction with the rest of the population, whereas G_i accounts for how strongly it influences the dynamics of other oscillators. Furthermore, in this formulation, the coupling scheme in Eq. (1) can be seen as a linear diffusive process between different groups with negligible spatially embedding effects [8,9]

Heterogeneity in the interaction patterns is introduced by dividing the population of N oscillators into two equally sized subpopulations, denoted by labels “1” and “2,” and characterized by two pairs of couplings (K_1, G_1) and (K_2, G_2) . All these constants are allowed to be positive or negative. Henceforth, we adopt the parametrization

$$K_{1,2} = K_0 \pm \frac{\Delta K}{2} \text{ and } G_{1,2} = G_0 \pm \frac{\Delta G}{2}, \quad (2)$$

where K_0, G_0 are the average in- and out-coupling and $\Delta K, \Delta G$ the respective mismatches. Furthermore, we speak of mixed interactions, if both positive and negative couplings are present, i.e., always when the condition $|\Delta K|/2 > K_0$ or $|\Delta G|/2 > G_0$ is satisfied.

Having defined the phases, the level of phase synchronization is measured by the order parameter $r = \langle |N^{-1} \sum_{j=1}^N \exp[i\phi_j^H(t)]| \rangle_t$, where $|\cdot|$ is the absolute value and $\langle \cdot \rangle_t$ denotes temporal average. Organizing the index labeling such that $j \in \{1, \dots, N/2\}$ refers to oscillators belonging to subpopulation 1 and $j \in \{N/2 + 1, \dots, N\}$ to subpopulation 2, it is also convenient to calculate the coherence within each group as $r_1 e^{i\Theta_1(t)} = (N/2)^{-1} \sum_{j=1}^{N/2} \exp[i\phi_j^H(t)]$ and analogously for $r_2 e^{i\Theta_2(t)}$, with $\Theta_{1,2}(t)$ being the corresponding mean phases. The phase lag $\delta(t)$ is then defined as the difference between the mean phases $\delta(t) = |\Theta_1(t) - \Theta_2(t)|$. Moreover, we calculate the wave speed as [11]

$$\Omega = \frac{1}{N} \sum_{j=1}^N \left\langle \frac{d\phi_j^H}{dt} \right\rangle_t. \quad (3)$$

The dynamical states are expressed as follows: (i) In the incoherent state, oscillators of both populations rotate independently so that $r_{1,2} \approx 0$. (ii) The oscillators are mutually attracted in the partial phase synchronized state yielding $r > 0$ and $\delta = 0$, which we denominate as “zero-lag sync.” (iii) A similar state to (ii) is attainable by system Eq. (1). Specifically, we refer to “lag sync” when partial global synchronization is achieved ($r > 0$) with $0 < \delta < \pi$ and in the absence of additional drift ($|\Omega - \omega_0| \simeq 0$). (iv) In the π state oscillators belonging to the same subpopulations are partially synchronized in phase ($r_{1,2} > 0$), while the mean phases are separated by $\delta = \pi$. (v) Finally, in the TW state, the phases drift with a frequency different from their intrinsic one yielding $|\Omega - \omega_0| > 0$. Figure 1 depicts snapshots of the trajectories of $N = 200$ weighted x -coupled Rössler oscillators [Eq. (1)]. Precisely, Fig. 1(a) shows a typical long-term configuration of a π state, where the populations are diametrically opposed. A typical lag sync state in the weighted x -coupled Rössler model is exemplified in Fig. 1(b), where the centroids of the two clusters are separated by a phase-lag $\delta < \pi$.

Before continuing, we should remark on the possibility of facing trajectories that escape the attractor after a period of transient chaos. Divergent trajectories arise due to the

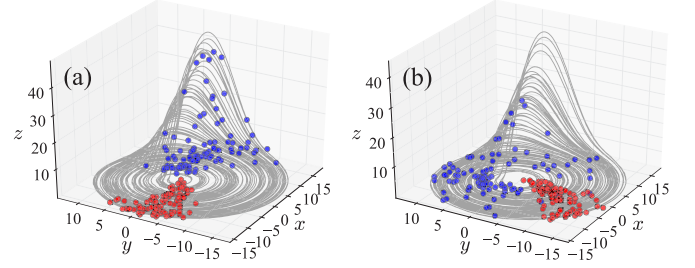


FIG. 1. Snapshots of the oscillators' trajectories of the weighted x -coupled Rössler model [Eq. (1)] with phase-coherent attractor ($a = 0.15$, $b = 0.2$, and $c = 10$). (a) π -state where $\delta = \pi$ with $K_0 = 0$ and $G_0 = 0$. (b) Lag sync state for which $\delta < \pi$ with $K_0 = 0$ and $G_0 = 0.2$. Gray line in the background corresponds to a trajectory of a randomly selected oscillator in order to depict typical dynamics. Remaining parameters: $N = 200$, $\Delta K = 0.1$, $\Delta G = 1$, and $D = 0.03$. See Supplemental Material [7] for the animated version of the figure and additional videos on other collective states.

dissipative character of the Rössler dynamics and are in general related to the conversion of a chaotic attractor into a nonattracting chaotic invariant set [12]. In order to show how this is manifested with weighted x -coupled Rössler oscillators in the absence of noise, Fig. 2 depicts a sample of the plane $G_0 \times K_0$ where trajectories that diverge to infinity are possible. For large values of $|G_0 K_0|$, divergences from the attractor are inevitable, but still a wide region with nondivergent orbits is observed. Noteworthy, the collective escapes are induced by the coupling, since isolated oscillators do not escape for the parameters considered here as shown in Ref. [12].

Taking Fig. 2 as a guide so that the parameters are chosen in a region free from escaping trajectories, in Fig. 3 we show the dependence of r , Ω , and δ as a function of the mean

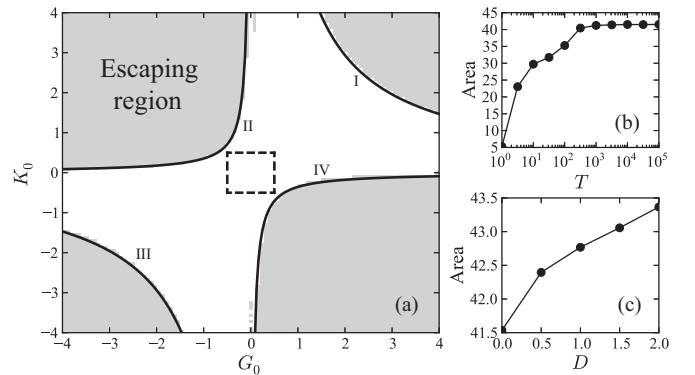


FIG. 2. (a) Map showing the regions with escaping trajectories for weighted x -coupled Rössler oscillators [Eq. (1)] in the phase-coherent regime considering $N = 500$, $D = 0$, $\Delta K = 0.1$, and $\Delta G = 1$. For each pair (G_0, K_0) the initial conditions $[x_i(0), y_i(0), z_i(0)] \forall i$ are randomly drawn according to a uniform distribution in the range $[-1, 1]$. Simulation time $T = 1 \times 10^5$. Solid black curves fitting the boundaries of the escaping region are obtained through the relation $\langle (KG) \rangle = c$, where $c_I = c_{III} = 11.8$ and $c_{II} = c_{IV} = -0.66$. Square at the center depicts the parameter region of Fig. 3. Area of escaping region as function of (b) simulation time T for fixed $D = 0$ and (c) as function of D for $T = 5 \times 10^3$.

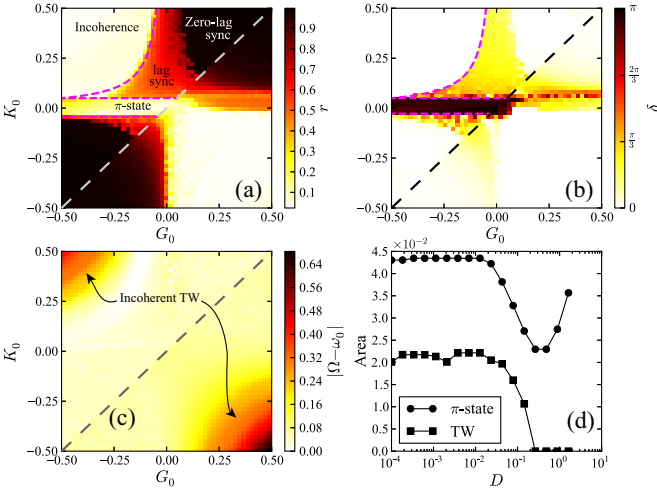


FIG. 3. Color maps showing (a) order parameter r , (b) phase lag δ , and (c) speed of phase wave $|\Omega - \omega_0|$ for weighted x -coupled Rössler oscillators [Eq. (1)] with phase-coherent (upper triangles) and funnel (lower triangles) attractors considering $D = 0.03$. (d) Area of TW state in which $|\Omega - \omega_0| \geq \varepsilon$, where we consider $\varepsilon = 0.3$; and π state for $\delta \geq \varepsilon$ with $\varepsilon = 3$. Remaining parameters: $\Delta K = 0.1$, $\Delta G = 1$, $N = 10^3$, 50×50 grid, simulation time $T = 5 \times 10^3$. Curves in the interface between incoherent and partial synchronized state for phase-coherent oscillators are given by the condition $2D_{\text{eff}} = \langle \langle KG \rangle \rangle$, here with $D_{\text{eff}} \approx 0$ [13]. Horizontal dashed lines correspond to the critical points $K_{0c} = \pm \Delta K/2$. See also Appendix A for diagrams with $\Delta K = \Delta G = 0$.

in- and out-couplings K_0 and G_0 , considering oscillators with phase-coherent, $(a, b, c) = (0.15, 0.2, 10)$, and funnel attractors, $(a, b, c) = (0.2, 0.2, 7)$. Making use of the symmetry of the maps with respect to the main diagonal, the results concerning the phase-coherent case are depicted in the upper triangles, whereas results regarding oscillators with funnel attractors are shown in the lower one. For all simulations regarding system Eq. (1) the long-time behavior of these quantities is calculated by averaging data between $t \in [4700, 5000]$ (see Appendix C). One notices the formation of a π -state in the region with $r \simeq 0$ [Fig. 3(a)] and $\delta \simeq \pi$ [Fig. 3(b)], which is surrounded by areas of partial synchronization with $0 < \delta < \pi$ for both types of attractor, yet less prominent for funnel (values of δ were set to 0 in the incoherent region, since Θ_1 and Θ_2 are undefined in this state). Interestingly, in the dynamics of phase oscillators under attractive and repulsive interactions, one would expect the emergence of a collective rhythm different from the natural frequency in the lag-sync region [4–6]. Differently weighted x -coupled Rössler oscillators subjected to weak noise intensities present $\Omega \simeq \omega_0$ for $0 \leq \delta \leq \pi$. On the other hand, for large negative values of $K_0 G_0$, significant values of $\Omega \neq \omega_0$ are observed in the incoherent region, as shown in Fig. 3(c). Correspondingly, we refer to this regime as incoherent TW.

Comparing Fig. 2 and Fig. 3(c), we see that $|\Omega - \omega_0|$ increases as one approaches the boundaries of the escaping region, predicting in a way the occurrence of divergent trajectories. We stress though that this behavior is not a result due to fluctuations, but an emergent phenomenon that

is yielded by the repulsive couplings considered in Eq. (1). Evidences of that can already be seen in Fig. 3. Note that deviations from the natural frequency ω_0 are restricted to TW areas, while the π and partially synchronized states are marked by an insignificant drift in the average frequency. Noteworthy, as shown in Fig. 3(c), for the phase-coherent case, the drift in the frequency reaches a minimum before the wave sets in, a phenomenon not observed for oscillators with funnel attractors. In Appendix A we further show that in system Eq. (1), global and strong repulsive interactions without coupling mismatches suffice to generate incoherent TWs, even in populations with size up to $N = 10^6$ (see also Appendix B).

The influence of noise is shown in Fig. 3(d) considering phase-coherent oscillators. Of particular interest is the non-monotonic dependence on the noise strength by the π -state area. On the other hand, large noise completely extinguishes the TW area. Similar results are found for oscillators with funnel attractors (not shown).

Despite the striking contrast between the nature of the TW states observed in these systems, it is still possible to quantitatively describe certain aspects of the dynamics of chaotic oscillators with theories developed in the context of pure phase oscillators. In Ref. [6] it is shown that the incoherent state ($r = 0$) loses its stability if $2D = \langle \langle KG \rangle \rangle$ is satisfied, where $\langle \langle \dots \rangle \rangle = \int dK' \int dG' \dots P(K', G')$ with $P(K', G')$ being the joint probability distribution of in- and out-couplings. Applying this condition in the weak noise case and considering the effective noise strength D_{eff} [13], we see that the result uncovered for the Kuramoto model predicts with great accuracy the boundaries of incoherent states. Another critical condition is found by noticing that at $K_{0c} = \pm \Delta K/2$ the oscillators of one of the populations are completely decoupled from the network. These conditions are depicted by the horizontal lines in Figs. 3(a) and 3(b), which encompass the π -state area.

III. PHASE-COUPLED RÖSSLER SYSTEMS

The levels of synchronization uncovered in the weighted x -coupled model share similarity with the ones found in ensembles of phase oscillators subjected to attractive and repulsive couplings [4–6]. However, careful inspection of the collective frequency diagrams [Fig. 3(c)] reveals that the dynamical states in these systems are in fact different. Precisely, TWs among weighted x -coupled chaotic oscillators are manifested only in the absence of coherent oscillations, while partial synchronization is crucial for the emergence of such states in coupled phase oscillators. This poses the question of whether incoherent TWs are induced by the intrinsic chaotic dynamics or by the type of coupling adopted in Eq. (1).

To shed light on this, we consider now a system of N Rössler systems with Kuramoto phase-couplings [14]. In the phase-coherent regime it is valid to separate the original system into phase and amplitude dynamics. For this purpose, one goes to cylindrical coordinates (ρ, ϕ, z) via $x(t) = \rho(t) \cos \phi(t)$ and $y(t) = \rho(t) \sin \phi(t)$. This results in $\dot{\rho} = a\rho \sin^2 \phi - z \cos \phi$, $\dot{\phi} = \omega + a \sin \phi \cos \phi + z/\rho \sin \phi$, $\dot{z} = b + \rho z \cos \phi - cz$. We then formulate the phase-coupled

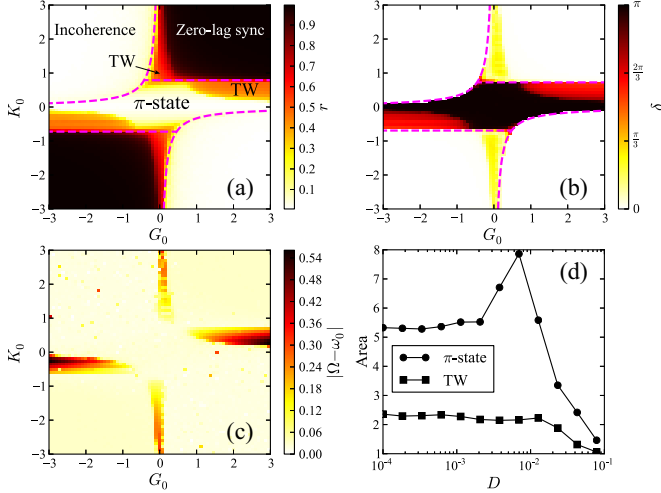


FIG. 4. Color maps for the phase-coupled model [Eq. (4)] showing (a) (total) order parameter r , (b) phase lag δ , (c) the spontaneous drift measured by $|\Omega - \omega_0|$, where Ω is the wave speed [Eq. (3)], and ω_0 the natural frequency. (d) Area of TW state in which $|\Omega - \omega_0| \geq \varepsilon$, where we consider $\varepsilon = 0.3$; and π state for $\delta \geq \varepsilon$ with $\varepsilon = 3$. For each point in the 70×70 grid, simulations are performed by evolving Eqs. (4) considering initial conditions $\rho_i(0)$ and $z_i(0)$ randomly chosen from uniform distribution $[-1, 1]$ and $\phi_i(0)$ uniformly distributed in the range $[-\pi, \pi]$. Other parameters: $N = 10^3$, $\Delta K = 1.45$, $\Delta G = 1$, $D = 0.03$, averaging over $t \in [4000, 5000]$ and $dt = 0.05$. Curves in the interface between incoherent and partial synchronized state are given by the condition $2D = \langle(KG)\rangle$. Horizontal dashed lines correspond to the critical points $K_{0c} = \pm \Delta K/2$. See Appendices D and E for additional diagrams of the phase-coupled model.

model as

$$\begin{aligned} \dot{\phi}_i &= \omega_0 + a \sin \phi_i \cos \phi_i + \frac{z_i}{\rho_i} \sin \phi_i \\ &+ \frac{K_i}{N} \sum_{j=1}^N G_j \sin(\phi_j - \phi_i) + \xi_i(t), \\ \dot{\rho}_i &= a \rho_i \sin^2 \phi_i - z_i \cos \phi_i, \\ \dot{z}_i &= b + \rho_i z_i \cos \phi_i - c z_i, \end{aligned} \quad (4)$$

$i = 1, \dots, N$. We consider here $a = 0.15$, $b = 0.2$, and $c = 10$ so that the oscillators are in the phase-coherent regime. The parametrization of the coupling strengths is again adopted as in Eq. (2).

We continue our analysis by getting a general view of the possible dynamical states of system Eqs. (4). In order to do so, Fig. 4 shows the simulation results of the total order parameter r , phase lag δ between the two populations, and the wave speed Ω , which are now calculated with respect to the phases defined in Eqs. (4). For each of the 70×70 points in the grid, system Eqs. (4) is numerically integrated by using the Heun's scheme with time step $dt = 0.05$ and considering a population of $N = 10^3$ oscillators for which the quantities of interest are averaged over $t \in [4000, 5000]$. By inspecting Figs. 4(a)–4(c), four well-defined states are uncovered, namely incoherent ($r_{1,2} \simeq 0$, $r \simeq 0$, and $|\Omega - \omega_0| \simeq 0$), zero-lag sync ($r \simeq 1$, $\delta \simeq 0$, and $\Omega \simeq 0$), π state ($r \simeq 0$, $\delta \simeq \pi$, and $|\Omega - \omega_0| \simeq 0$) and

TW ($r > 0$, $0 < \delta < \pi$, and $\Omega \neq \omega_0$). Interestingly, in contrast to the weighted x -coupled model, TWs are not observed in the presence of incoherence if the oscillators are coupled only through their phase angles. Another difference between the coupling models is the fact that the areas of the π state behave in a different manner as a function of the noise strength D [compare Fig. 3(d) with Fig. 4(d)].

Let us give a brief overview on the effects of noise observed here. (i) Noise enlarges the region of collective escape, but only in a surprisingly slight way; see Fig. 2(c). (ii) In the x -coupled case the incoherent TWs disappear at a critical noise intensity, while the π states are favored both by low and high noise intensities; see Fig. 3(d). (iii) In the phase-coupled case TW states again disappear at large noise intensities, but π states are most prominent at an intermediate noise intensity; see Fig. 4(d).

IV. CONCLUSION

The coexistence of chaotic, noisy, coherent, and incoherent collective dynamics can be observed in many real-world systems [15]. Here we provided the first evidence that TWs, i.e., states in which a new rhythm among the oscillators different from the natural frequency emerges, are also attainable in the phase synchronization of chaotic Rössler oscillators. Two coupling formulations were analyzed. First, we connected the oscillators via a weighted linear coupling in the x coordinates, and we considered thereby both chaotic regimes, the phase-coherent and the funnel attractor scenario. In both cases, novel incoherent TWs were observed, which are characterized by the appearance of a new rhythm of oscillation without global synchronization. Second, we considered purely phase-coupled Rössler systems in the phase-coherent regime. In this second case, TWs were only found in conjunction with phase synchronization, namely in regions where the phase lag between the two synchronized groups fulfills $0 < \delta < \pi$.

Our findings on the one hand reinforce the potential that phase models possess in describing the dynamics of higher-dimensional systems, but on the other hand they also highlight the limitations of phase models regarding other kinds of couplings, such as in the prediction of incoherent TWs. More precisely, in ensembles of phase oscillators and in phase-coupled Rössler systems introduced here, TWs are accessible if the subpopulations are separated by a phase lag whose existence is only possible under partial synchronization. However, the assumption that partial synchronization is necessary for the occurrence of TWs ceases to be true for diffusively coupled Rössler systems. Therefore, although the routes to phase synchronization of chaotic oscillators are qualitatively explained by phase models [1], incoherent TWs cannot be described by such a decoupling of phases and amplitudes, since this phenomenon has no counterpart in ensembles of phase oscillators. This fact demonstrates a significant difference in the collective states observed in the phase dynamics of chaotic and phase oscillator systems.

It remains as a future research to thoroughly determine the boundaries of the escaping region in Fig. 2 as well as the investigation of whether other chaotic systems also exhibit similar dynamical patterns as the ones described here. Likewise promising is the investigation of chimeralike

states [16] in the context of attractive and repulsive couplings. We further expect to observe the aforementioned states in real experiments with, for instance, electrochemical Belousov-Zhabotinsky oscillators [17].

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APPENDIX A: INCOHERENT TRAVELING WAVES IN THE ABSENCE OF COUPLING MISMATCH

In the main text, it was shown that incoherent TWs emerge for large negative values of $K_0 G_0$ in the presence of in- and out-coupling mismatches. However, this poses the question about whether such states are also attainable in the absence of coupling mismatches between the oscillators. Figure 5 shows the order parameter r and the drift in the frequencies $|\Omega - \omega_0|$ of weighted x -coupled Rössler oscillators [Eq. (1)] with phase-coherent and funnel attractors as function of K_0 and G_0 considering $\Delta K = \Delta G = 0$. Again, as in the case where $\Delta K \neq 0$, $\Delta G \neq 0$, TWs are found in regions with sufficiently negative values of $K_0 G_0$. Interestingly, this shows that the presence of a phase-lag between two populations is not a

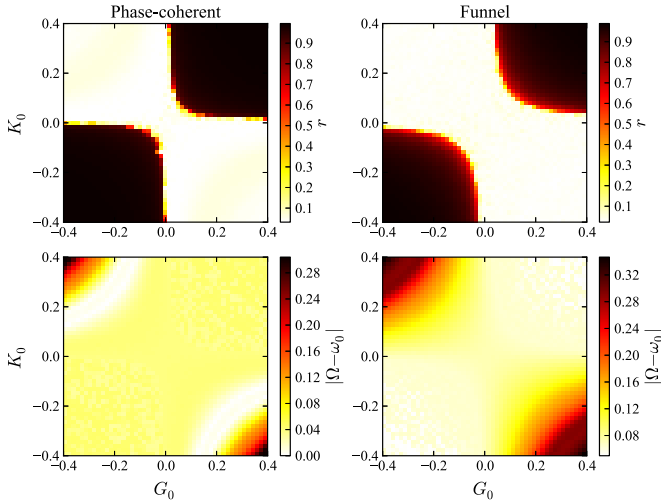


FIG. 5. Color maps for weighted x -coupled Rössler oscillators with phase-coherent ($a = 0.15$, $b = 0.2$, and $c = 10$) and funnel attractors ($a = 0.2$, $b = 0.2$, and $c = 7$) in the absence of coupling mismatches, i.e., $\Delta K = 0$, $\Delta G = 0$. Parameters: $N = 10^3$, $dt = 0.05$. For each (G_0, K_0) in the 50×50 grid, the quantities are averaged over $t \in [4700, 5000]$. Initial conditions $(x_i(0), y_i(0), z_i(0))$ are selected from an uniform distribution in the range $[-1, 1]$.

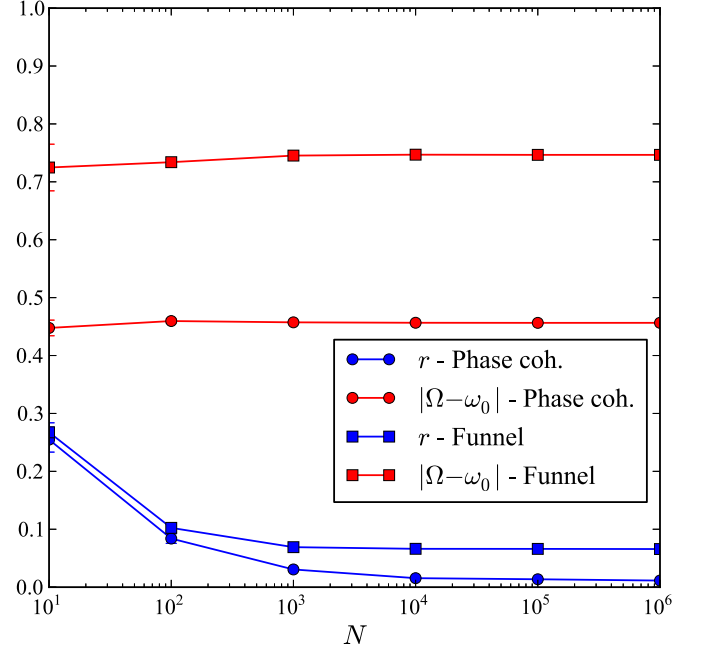


FIG. 6. Dependence of incoherent TWs on the system size for $K_0 = 0.5$, $G_0 = -0.5$, $\Delta K = 0.1$, $\Delta G = 1$, and $D = 0.03$. Each point is an average over 100 different realizations. Size of error bars is smaller than the dots' sizes.

necessary condition for the emergence of a spontaneous drift different from the ensemble's mean frequency, contrary to what is observed in the collective dynamics of conformists and contrarians phase oscillators [4,6]. In fact, as shown in Ref. [6], not only the condition $\delta \neq 0$, but also asymmetric coupling is required for the existence of TWs in ensembles of phase oscillators. This suggests that asymmetric coupling is not a necessary condition neither for the observation of incoherent TWs.

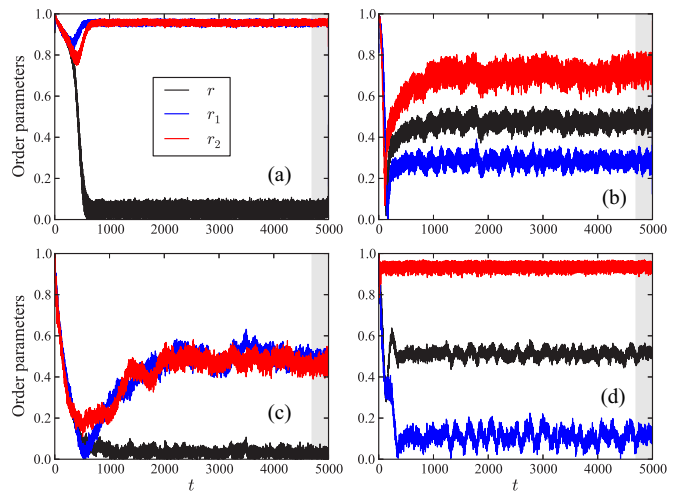


FIG. 7. Examples of time evolution of order parameters for phase-coherent oscillators in (a) π state ($K_0 = G_0 = 0$), (b) lag sync ($K_0 = 0.25$, $G_0 = -0.1$); and oscillators with funnel attractor also in (c) π state ($K_0 = G_0 = 0$) and (d) lag sync ($K_0 = 0.03$, $G_0 = 0.5$). Other parameters: $\Delta K = 0.1$, $\Delta G = 1$, $D = 0.03$, and $N = 10^3$. Shaded area corresponds to the time-averaging window considered in Fig. 3.

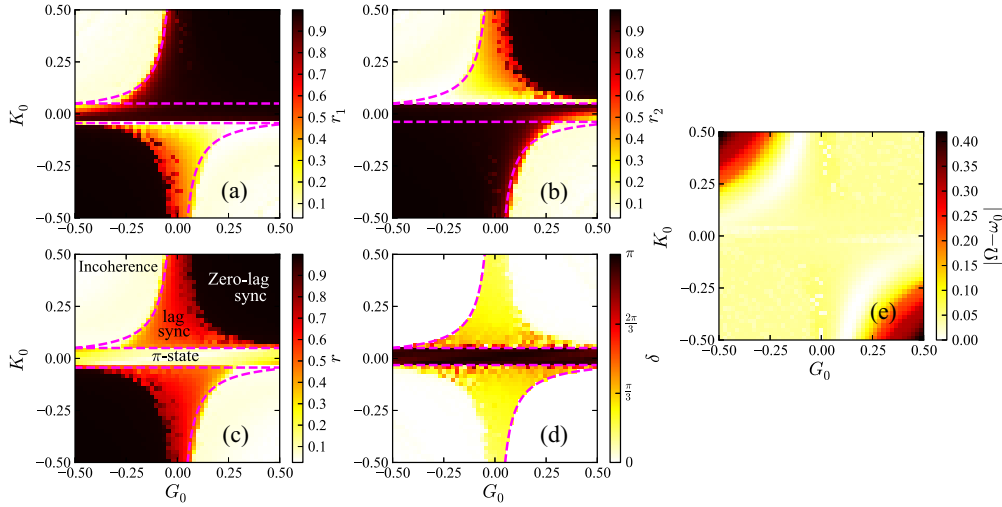


FIG. 8. Color maps for weighted x -coupled Rössler oscillators with phase-coherent attractor ($a = 0.15$, $b = 0.2$ and $c = 10$). (a) r_1 , (b) r_2 , (c) r , (d) $|\Omega - \omega_0|$, and (e) δ . Remaining parameters: $\Delta K = 0.1$, $\Delta G = 1$, $D = 0.03$, and $N = 10^3$. Notice in panel (d) that the white upper left and lower right regions are undefined since the phase lag is undefined. On the other hand, the frequency difference (e) defined via the Hilbert transform [Eq. (3)] indicates a strict acceleration.

Note that here only a single coupling strength exists, namely $K_0 G_0$. However, in order to better compare with the main text, we show the results in the $G_0 \times K_0$ plane.

APPENDIX B: DEPENDENCE OF INCOHERENT TWS ON SYSTEM SIZE

Another pertinent issue in our study is the verification of whether incoherent TWs persist in larger populations. In order to address this issue, in Fig. 6 we report $|\Omega - \omega_0|$ and r as function of the system size N considering $(G_0, K_0) = (-0.5, 0.5)$, parameters for which extreme values of $|\Omega - \omega_0|$ are observed in Fig. 3. Interestingly, no sensitive dependence on the number of oscillators is found for the magnitude of the TWs, showing that the phenomenon is indeed not a consequence of finite-size fluctuations.

APPENDIX C: EVOLUTION OF ORDER PARAMETERS

In this section we report on the relaxation dynamics of weighted x -coupled Rössler oscillators. Figure 7 shows typical evolutions of the order parameters observed in lag sync and π states. Similar results are obtained for zero-lag sync and incoherent states as well as for phase-coupled oscillators.

APPENDIX D: LOCAL ORDER PARAMETERS

In order to also characterize the dynamics within the subpopulations, in this section we present diagrams for the local order parameters $r_{1,2}$ for weighted x -coupled and phase-coupled models, shown in Figs. 8 and 9, respectively. We repeat the respective maps for variables r , δ , and Ω shown in Figs. 3 and 4. Noteworthy, we remark that one does not need to rely on parameters r_1 and r_2 when detecting TWs. The latter states are solely described by the emergence of a new frequency different

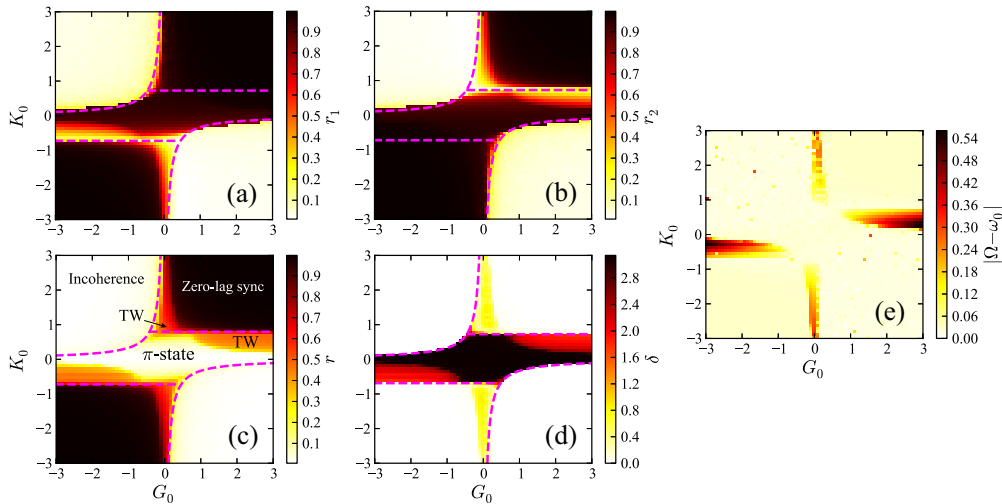


FIG. 9. Color maps for phase-coupled Rössler oscillators with phase-coherent attractor ($a = 0.15$, $b = 0.2$, and $c = 10$) for $\Delta K = 1.45$, $\Delta G = 1$, $D = 0.03$, and $N = 10^3$. Concerning (d), see the remark on the phase lag in Fig 8.

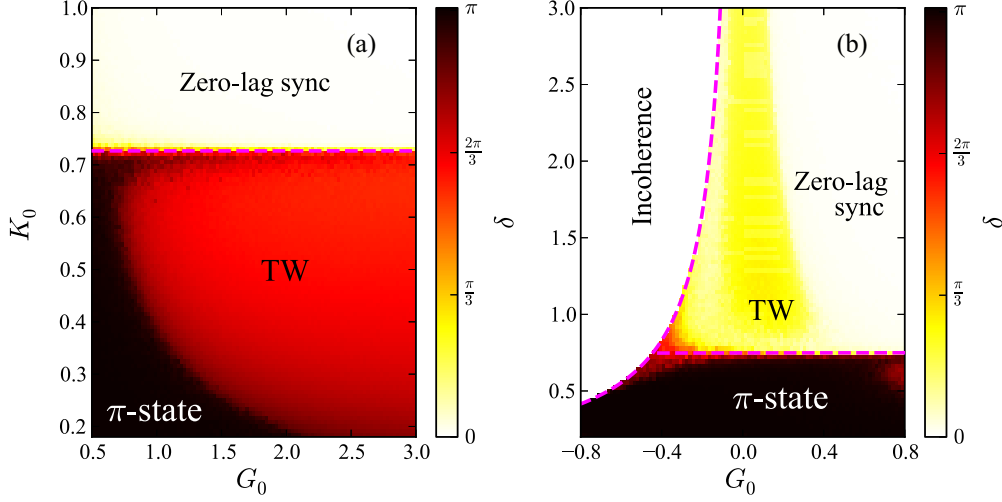


FIG. 10. The absolute value of the difference between the mean phases of the two populations, δ , in dependence on the mean in- and out-coupling strengths, K_0 and G_0 , respectively. The two panels show zoomed regions from Fig. 4, with different transition scenarios into the TW states. In (a) the TW region is surrounded by zero-lag states, whereas in (b) the TW states are surrounded by π states. Parameters: $N = 10^3$, $\Delta K = 1.45$, $\Delta G = 1$, $D = 0.03$, $dt = 0.05$. For each (G_0, K_0) in the 100×100 grid, the quantities averaged over $t \in [4000, 5000]$. Dashed magenta line in panel (a) marking the interface between Incoherence and Zero-lag sync is obtained by the critical condition $2D = \langle \langle KG \rangle \rangle$, where $\langle \langle \dots \rangle \rangle = \int dK' \int dG' \dots P(K', G')$ and $P(K, G)$ is the joint probability distribution of in- and out-couplings. Horizontal dashed lines in both panels correspond to the critical points $K_{0c} = \pm \Delta K / 2$.

from the natural one, i.e., when the condition $|\Omega - \omega_0| > 0$ is fulfilled. The consideration of other parameters is necessary only when distinguishing incoherent from coherent TWs.

APPENDIX E: ADDITIONAL DIAGRAMS FOR THE PHASE-COUPLED MODEL

Figure 10 shows zoomed color maps around the TW regions of Fig. 4 related to the phase-coupled model [Eq. (4)]. Precisely, Fig. 10(a) depicts the TW state surrounded by the π -state region, whereas Fig. 10(b) highlights the TW state reached via the zero-lag state. We recall that the emergence of TW and π states as observed in Fig. 10 in the phase-coupled Rössler model bears an interesting similarity with the stochastic Kuramoto model in Ref. [6], despite the fact that in the former the phases evolution explicitly depend on the chaotic amplitudes.

APPENDIX F: COMPARISON BETWEEN LOCKING AND MEAN ENSEMBLE FREQUENCIES

Traveling phase waves are generally detectable either via the mean ensemble frequency Ω [Eq. (3)] or in terms of the locking frequency Θ . Here we compare the performance of both measures when evaluating the TW regions for the phase-coupled Rössler model. Figure 11(a) reproduces the color map of Fig. 4(c), while Fig. 11(b) shows the collective drifts in the frequency measured by $|\langle \dot{\Theta} \rangle_t - \omega_0|$. As it is seen, the same patterns are uncovered with the particular difference that higher values for the latter measure are obtained. This fact is expected, since only oscillators synchronized with the mean-field contribute in the sum of Eq. (3) [6]. In Fig. 12 we compare in more detail the measures considering a small region of the parameter space in Fig. 11 for fixed G_0 . As shown, the onset of the TW region is equally captured by both definitions. Finally, for the sake of comparison between

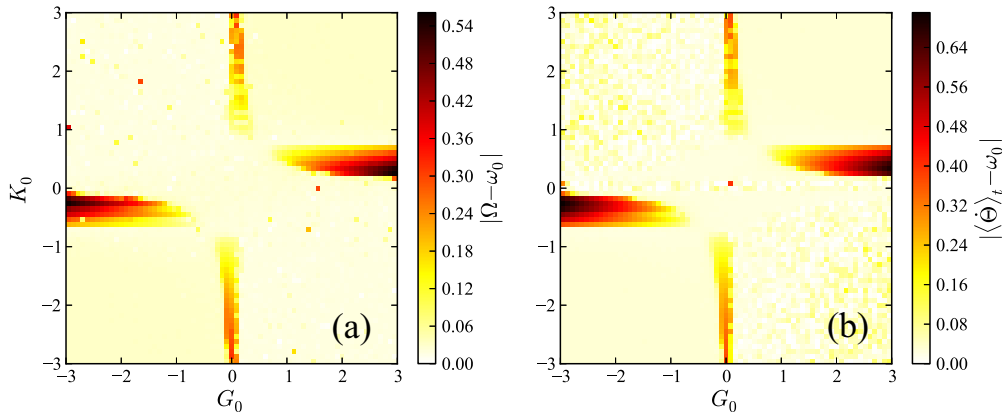


FIG. 11. Color maps for (a) $|\Omega - \omega_0|$ and (b) $|\langle \dot{\Theta} \rangle_t - \omega_0|$ of phase-coupled Rössler oscillators. Parameters: $N = 10^3$, $\Delta K = 1.45$, $\Delta G = 1$, $D = 0.03$, averaging over $t \in [4000, 5000]$ and $dt = 0.05$.

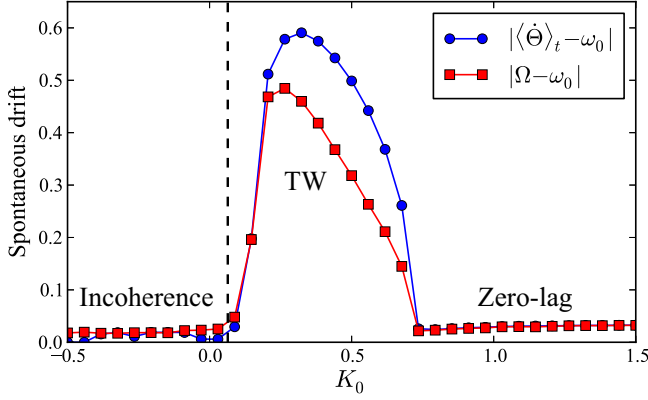


FIG. 12. Spontaneous drift measured by locking frequency $\langle \dot{\Theta} \rangle_t$, and mean ensemble frequency Ω . Parameters: $N = 10^4$, $\Delta K = 1.45$, $\Delta G = 1$, $G_0 = 2.5$, $D = 0.03$, averaging over $t \in [4000, 5000]$ and $dt = 0.05$.

coherent and incoherent TWs, Fig. 13 depicts, for fixed K_0 , the variation of $|\Omega - \omega_0|$ in regions of incoherence and zero-lag synchronization considering oscillators with funnel attractors.

APPENDIX G: VIDEOS DESCRIPTION

(1) `video1_weighted_coupled_model.mp4`: Animated version of Fig. 1. The video displays the trajectories of $N = 200$ of weighted x -coupled Rössler oscillators with phase-coherent attractors ($a = 0.15$, $b = 0.2$, and $c = 10$). Panel (a) shows a π state, where $\delta = \pi$ with $K_0 = 0$ and $G_0 = 0$. (b) Lag sync state for which $\delta < \pi$ with $K_0 = 0$ and $G_0 = 0.2$. Remaining parameters: $\Delta K = 0.1$, $\Delta G = 1$, and $D = 0.03$.

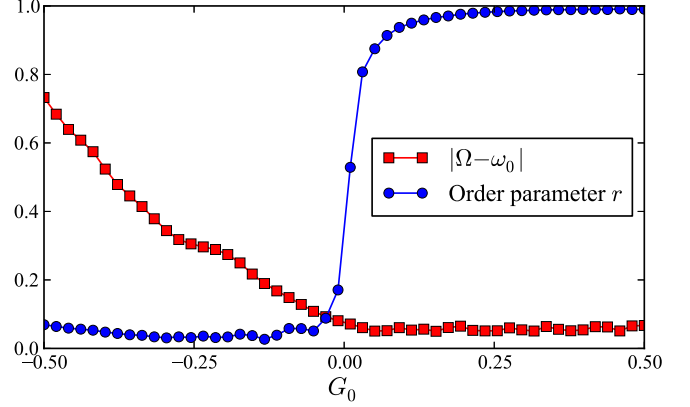


FIG. 13. Order parameter r and mean ensemble frequency Ω as functions of G_0 in regions of incoherence and zero-lag synchronization for oscillators with funnel attractors. Parameters: $N = 10^3$, $\Delta K = 0.1$, $\Delta G = 1$, $K_0 = 0.5$, and $D = 0.03$.

(2) `video2_phase_coupled_model.mp4`: Video displaying the trajectories of $N = 200$ phase-coupled Rössler oscillators. (a) π -state where $\delta = \pi$ with $K_0 = 0$ and $G_0 = 0$. (b) TW state where $\delta < \pi$ and $|\Omega - \omega_0| > 0$ with $K_0 = 0.3$ and $G_0 = 3.0$. Remaining parameters: $\Delta K = 1.45$, $\Delta G = 1$, and $D = 0.03$.

(3) `video3_incoherent_TWs.mp4`: Video showing an incoherent TW state of $N = 200$ weighted x -coupled Rössler oscillators with funnel attractors. Other parameters: $K_0 = 0.5$, $G_0 = -0.5$, $\Delta K = 0.1$, $\Delta G = 1$, and $D = 0.03$.

In all videos, the gray lines in the background correspond to a trajectory of a randomly selected oscillator in order to depict typical local dynamics. Also in all videos, the green dot stands for a completely uncoupled oscillator with same natural frequency $\omega_0 = 1$ as all the others to serve as reference.

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